

9 Cryptography

9.3 Public Key Cryptography

9.3.1 Modular Arithmetic

Definition 9.3.1 The **modulus** is another name for the remainder after division.

Example 9.3.2 Compute

1. $15 \bmod 7$
2. $125 \bmod 43$
3. $-12 \bmod 7$
4. $2^7 \bmod 5$

Note that we will always write our final answer as a natural number between 0 and $n - 1$.

Example 9.3.3 Your turn! Compute the following

1. $123 \bmod 19$
2. $-923 \bmod 134$
3. $-124 \bmod 543$
4. $1045 \bmod 521$
5. $8324 \bmod 358$

Theorem 9.3.4 (*Basic facts about modular arithmetic*) when n is a natural number, the following are true:

1. $(a + b) \bmod n = (a \bmod n) + (b \bmod n)$
2. $(ab) \bmod n = (a \bmod n)(b \bmod n)$
3. $a^b \bmod n = (a \bmod n)^b \bmod n$

Example 9.3.5 Compute the following:

1. $128 \cdot 8935 \bmod 5$
2. $4356^{2532} \bmod 4355$
3. $4354^{2532} \bmod 4355$

Example 9.3.6 Compute the following:

1. $47^8 \bmod 49$
2. $65^5 \bmod 35$

Example 9.3.7 Your turn! Compute the following:

1. $12^8 \bmod 4$
2. $12^8 \bmod 5$
3. $5^{12} \bmod 4$
4. $98^{123} \bmod 99$

What else can we do with modular arithmetic?

Example 9.3.8 Compute each of the following:

1. $3^{10} \pmod{22}$
2. $5^{10} \pmod{22}$
3. $21^{10} \pmod{22}$

Talk briefly about Euler's Theorem:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

when $\gcd(a, n) = 1$.

Definition 9.3.9 Euler's totient function ϕ is defined as follows: If n is a positive whole number, and $p_1, p_2, \dots, p_n$ are all the prime numbers dividing n , then

$$\phi(n) = n \left(1 - \frac{1}{p_1}\right) \left(1 - \frac{1}{p_2}\right) \cdots \left(1 - \frac{1}{p_n}\right)$$

Example 9.3.10 Compute the following:

1. $\phi(27)$
2. $\phi(19)$
3. $\phi(42)$
4. $\phi(124)$

Example 9.3.11 So how might we compute something like the following:

$$128^{1324} \pmod{34}$$

16

Example 9.3.12 Your turn! Compute

$$415^{1521} \bmod 27.$$